

Article

Small-Data Neural Computing Outperforms RSM: Low-Cost Smart Optimization in Injection Molding

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Featured Application

The proposed small-data neural computing framework, integrating Taguchi methods with Moldex3D digital twin simulations, provides an efficient and low-cost optimization tool for plastic injection molding. This virtual–physical integration strategy reduces the number of physical trials by 96%, corresponding to an estimated manufacturing cost reduction of 15–25%. In addition, precise control of shrinkage (3.079% with a relative error of 0.19%) significantly reduces the scrap rate.

Abstract

In smart manufacturing, the injection molding industry faces a “data scarce environment” due to prohibitive physical trial costs. Processing recycled polypropylene (rPP) exacerbates this challenge, as traditional response surface methodology (RSM) fails to capture complex non-linear rheological behaviors induced by material variability. This study proposes a “domain-knowledge guided data augmentation framework,” integrating Taguchi experimental data (L_{25}) with Moldex3D digital twin simulations to construct a 300-sample hybrid dataset. A back-propagation neural network (BPNN) with L_2 regularization was employed for small-sample learning, providing a continuous differentiable physical mapping. To rigorously prevent neighborhood data leakage, the model was evaluated via a strict nested group-based 5-fold cross-validation. Particle swarm optimization (PSO) was coupled to overcome the local minima of gradient descent. Comparative analysis demonstrates that BPNN significantly outperforms both traditional RSM and a newly introduced Random Forest (RF) baseline, achieving a testing mean squared error (MSE) of 0.001 (± 0.0002) and a testing R^2 of 0.95. PSO minimized the shrinkage rate to 3.079%, validated via Moldex3D digital twin simulation with a 0.19% relative error. Synergizing virtual–physical integration with robust neural computing enables superior process control precision in small-data regimes, offering small and medium-sized enterprises (SMEs) a cost-effective pathway for smart optimization.



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Keywords: small-data neural computing; data augmentation; response surface methodology (RSM); recycled polypropylene; particle swarm optimization (PSO)

1. Introduction

1.1. The “Small Data” Challenge and Multi-Physics Sensing Limitations in Smart Manufacturing

With the advancement of Industry 4.0, data-driven approaches have become a cornerstone for enhancing quality and efficiency in injection molding [1]. While deep learning

models have demonstrated exceptional capabilities in feature extraction and non-linear mapping, their performance is often contingent upon large-scale labeled datasets [2]. However, in practical injection molding scenarios, acquiring “Big Data” presents a severe economic and technical barrier.

The injection molding process is inherently governed by complex multi-physics coupling, including rapid phase transitions, non-linear rheological variations, and dynamic heat transfer between the polymer melt and the mold cavity [3,4]. Capturing these temporal dynamic features in real-time requires the deployment of high-resolution, invasive in-mold sensors. Yet, the extreme thermo-mechanical environment (e.g., high injection pressures and rapid thermal cycles) often leads to sensor degradation, prohibitive installation costs, and signal-to-noise ratio issues in practical shop-floor operations [5,6].

Furthermore, each physical trial-and-error iteration incurs significant machine downtime, energy consumption, and material waste—particularly when utilizing high-cost virgin materials or property-unstable recycled polypropylene (Recycled PP). The unpredictable batch-to-batch variability of recycled materials further complicates the acquisition of stable sensor data [7]. Consequently, constructing a massive, high-fidelity dataset via thousands of physical experiments to capture the complete spatio-temporal dynamics is economically and technically impractical.

Manufacturing industries are thus confronted with a typical “Data-Scarce Environment”. The critical challenge in current smart manufacturing is how to construct a predictive model with high generalization capability using a strictly limited number of experimental samples (Small Data). Failure to overcome this data bottleneck will hinder the practical deployment of artificial intelligence in precision manufacturing.

1.2. The Mathematical Bottleneck of Traditional RSM in Capturing Polymer Crystallization Kinetics

For decades, Response Surface Methodology (RSM) has been the industry standard for process parameter optimization. Fundamentally, RSM constructs a continuous energy landscape relying on second-order polynomial regression, which intrinsically assumes a smooth and gentle curvature among interacting variables [8]. While this simplified quadratic approximation may suffice for amorphous plastics with stable PVT (Pressure-Volume-Temperature) trajectories, its predictive accuracy may significantly decrease when applied to semi-crystalline polymers like recycled polypropylene (rPP).

From the perspective of polymer physics, the cooling phase of rPP is governed by highly complex crystallization kinetics and thermo-rheological coupling [9]. Under extreme multi-physics boundary conditions—such as severe temperature gradients between the polymer melt and the mold cavity—the material undergoes rapid and drastic phase transitions. These transitions induce abrupt non-linear volumetric shrinkage and localized rheological instabilities [4]. Consequently, the interaction effects between parameters (e.g., melt temperature and packing pressure) create a highly “rugged” response landscape characterized by sharp curvature changes and multiple local optima [10].

Traditional RSM mathematically forces a “smoothing” effect over these localized thermodynamic anomalies, which often limits its prediction accuracy under extreme boundary conditions [11]. This inherent statistical limitation proves that quadratic polynomials lack the degrees of freedom required to resolve complex spatial non-linear thermo-rheological interactions. Therefore, to achieve high-precision quality control in data-scarce environments, there is a strong theoretical motivation to transition beyond linear statistical approximations and transition toward robust neural computing architectures—such as a Back-Propagation Neural Network (BPNN) utilizing ReLU activation functions—to establish a highly non-linear, continuous differentiable spatial mapping.

1.3. Research Gap and Contributions in Data-Scarce Environments

In recent years, the frontier of smart injection molding has gradually shifted towards capturing “temporal dynamic features” (e.g., the in-mold cavity pressure and temperature decay curves over time) using advanced time-series deep learning models such as Long Short-Term Memory (LSTM) networks or 1D Convolutional Neural Networks (1D-CNN) [12,13]. While these dynamic models exhibit exceptional predictive capabilities for localized defects, a critical academic critique must be raised: their performance is inherently predicated upon the availability of massive, high-fidelity, and continuous temporal datasets. In practical shop-floor operations, acquiring such dynamic “Big Data” through high-resolution invasive sensors is economically and technically prohibitive for most Small and Medium-sized Enterprises (SMEs), cementing the reality of a “data-scarce environment” [14].

Consequently, a significant research gap persists: how can we achieve high-precision optimization and establish a robust algorithmic foundation when dynamic big data is inaccessible, particularly when processing highly variable recycled polypropylene (rPP)?

To bridge this substantial gap, this study proposes a “Domain-Knowledge Guided Data Augmentation” framework. Instead of prematurely deploying data-hungry dynamic models, we synergized limited physical trials (Taguchi $L_{25}(5^4)$) with Moldex3D digital twin simulations to construct a 300-sample hybrid dataset. We successfully deployed an L_2 -regularized BPNN (4-10-1 topology) coupled with Particle Swarm Optimization (PSO). The core contributions of this study are twofold:

1. Superior Non-linear Spatial Mapping in Small-Data Regimes: We demonstrate that the BPNN provides a continuous differentiable spatial mapping, successfully capturing the highly non-linear thermo-rheological interactions of rPP that traditional statistical methods (RSM) fundamentally fail to resolve.
2. Establishing the Algorithmic Foundation for Future Adaptive Control: By achieving a highly accurate static parameter spatial mapping under severe data constraints, this framework serves as the indispensable algorithmic steppingstone. It proves the viability of low-cost neural computing and paves the way for future real-time dynamic adaptive control systems that will eventually integrate temporal dynamic features.

2. Methodology: From Statistical Design to Neural Computing

Based on the comprehensive research framework illustrated in Figure 1, the subsequent sections present a comparative analysis of the BPNN surrogate model against the RSM baseline, followed by the global process optimization of rPP shrinkage.

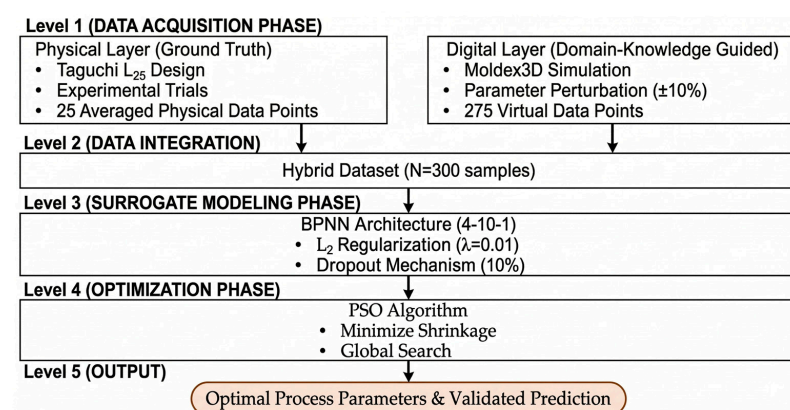


Figure 1. Research Framework Flowchart: Synergizing virtual-physical integration (Taguchi-Moldex3D) with a robustly regularized BPNN-PSO for low-cost process optimization in data-scarce environments.

2.1. Data Acquisition Strategy: Domain-Knowledge Guided Augmentation

Given the high cost of physical trials in injection molding, this study employs a hybrid data acquisition strategy that integrates physical experimentation with physics-based simulation. This approach, termed “Domain-Knowledge Guided Data Augmentation,” aims to construct a comprehensive feature space that captures the nonlinear rheological behaviors of the material under limited experimental resources [15].

- **Physical Layer (Ground Truth):** The core dataset was derived from a rigorous experimental design using the Taguchi $L_{25}(5^4)$ orthogonal array. Four key process parameters were varied across five levels: melt temperature (180–220 °C), mold temperature (20–50 °C), packing pressure (50–150 MPa), and cooling time (10–30 s). This five-level configuration was strategically designed to initially capture high-order non-linear features while averting an exponential increase in physical trial costs. Each experimental condition was replicated five times to mitigate the stochastic noise of the recycled material and ensure statistical reliability, resulting in 125 physical data points. The run-to-run variability across these five physical replicates was statistically evaluated, yielding an average standard deviation of $\pm 0.015\%$. This metric quantitatively confirms the inherent rheological fluctuations and batch variability of the recycled PP. These 125 physical trials were averaged to form 25 robust ground-truth points, filtering out the stochastic noise of the recycled material [16].
- **Digital Layer (Data Augmentation):** To overcome the “small data” limitation inherent in manufacturing datasets, we utilized Moldex3D 2025 CAE software to generate an additional 275 virtual data points. Crucially, the $\pm 10\%$ parameter perturbation is not a mere mathematical interpolation or random numeric augmentation. Instead, these perturbed boundary conditions were re-inputted into the Moldex3D solver, where they were strictly constrained by the physical conservation laws and governing equations of polymer flow and heat transfer. This ensures that the generated non-linear thermo-rheological responses are physically realistic and independent, proving that the augmented dataset allows the network to learn the true underlying physical mechanisms rather than simply memorizing mathematical perturbation rules [9]. This augmented dataset (total $N = 300$) bridges the gaps in the experimental design space, allowing the neural network to learn the underlying physics rather than merely memorizing the sparse experimental points.
- **Material Context:** The material used was Polypropylene (PP-H703) blended with 50% recycled PP. The incorporation of recycled material introduces significant variability in viscosity and crystallization kinetics due to molecular weight degradation [7]. This material complexity necessitates a nonlinear modeling approach capable of capturing dynamic property fluctuations that traditional linear models often fail to address.

2.2. The Small-Data BPNN Architecture with Regularization

In data-scarce scenarios, simple machine learning methods such as linear regression, Decision Trees, or Random Forests are commonly employed. However, this study deliberately eschews tree-based models in favor of a Back-Propagation Neural Network (BPNN). The core rationale lies in the continuity of physical mechanisms. Tree-based models inherently perform a “step-wise” partitioning of the feature space. Yet, the shrinkage behavior of PP during injection molding is a continuous thermodynamic and rheological process, deeply coupled by temperature gradients and pressure transfer. The predictive surfaces generated by tree models are discontinuous and fail to authentically reflect the smooth, gradual variations of physical quantities within the mold cavity. In contrast, a BPNN equipped with a non-linear activation function, such as the Rectified Linear Unit (ReLU),

provides a highly non-linear continuous mapping, thereby finely capturing the intense non-linear interactions between melt temperature and mold temperature.

The network topology adopts a three-layer architecture with a single hidden layer (4-10-1): 4 input neurons corresponding to the process parameters, 10 hidden neurons, and 1 output neuron. The number of hidden neurons was systematically determined through a Grid Search hyperparameter tuning process integrated with a strict nested group-based 5-fold cross-validation. Specifically, the inner loop of this nested procedure was utilized to perform the Grid Search for optimal hyperparameters (e.g., hidden neurons) using the training subsets. Simultaneously, the outer loop was employed to evaluate the model's final performance metrics (such as MSE and R^2) on the completely unseen testing folds. Given that 275 simulated points were generated via perturbations around 25 foundational physical experiments, standard random splitting would inevitably lead to neighborhood leakage (i.e., the model memorizing the localized simulation space of a physical point). To rigorously prevent this, the dataset was grouped by the 25 physical cases. During the splitting process, all simulated data points generated from the same physical experiment were strictly tied together and assigned to the identical fold. Consequently, the testing set evaluated at each fold represented entirely unseen physical operational regions, ensuring the strict independence of the validation and testing phases. In a small-data regime ($N = 300$), this group-based splitting, combined with our shallow architecture, prevents overfitting and the curse of dimensionality. Our grid search confirmed that a single hidden layer with 10 neurons, combined with L_2 regularization, achieves an optimal balance between computational cost and model fitting capability. This shallow yet robust architecture is entirely sufficient to accurately extract the complex non-linear spatial features of recycled PP without inducing model collapse.

However, training deep learning models on a small dataset ($N = 300$) poses a significant risk of overfitting, where the model easily memorizes noise rather than learning generalizable physical patterns. To simplify the model and capture core variables, we implemented the following robust regularization strategies:

1. L_2 Regularization (Weight Decay, $\lambda = 0.01$): To penalize large weights and prevent overfitting, the cost function $J(\theta)$ is modified by adding a regularization term, as shown in Equation (1):

$$J(\theta) = \frac{1}{m} \sum_{i=1}^m L(\hat{y}^{(i)}, y^{(i)}) + \frac{\lambda}{2m} \sum_{j=1}^n \theta_j^2 \quad (1)$$

2. Here, L is the loss function, $\hat{y}^{(i)}$ and $y^{(i)}$ are the predicted and actual shrinkage values of the i -th sample, respectively, m is the total number of training samples, n is the total number of network weights, λ is the regularization parameter, and θ_j represents the individual weights. Dropout Mechanism (10%): A dropout rate of 0.1 was introduced, randomly deactivating 10% of the neurons during each training iteration. This technique forces the network to learn redundant representations of key features, enhancing its robustness against the inherent variability of recycled materials.

Why choose L_2 over L_1 regularization? From a physical perspective, L_1 regularization induces “sparsity” in the weight matrix, forcefully driving certain variable weights to zero. However, in injection molding, melt temperature, mold temperature, packing pressure, and cooling time all contribute indispensably to the physical boundary conditions governing shrinkage. Therefore, L_2 regularization smoothly penalizes and shrinks all weights without eliminating them. This suppresses the model's oversensitivity to singular noise in small datasets while fully preserving the holistic physical dynamics of multivariate coupling.

The aforementioned hyperparameters ($\lambda = 0.01$, Dropout = 0.1) were not randomly selected but determined through a Grid Search combined with cross-validation, ensuring optimal generalization capability under small-data conditions.

The complete architecture of the proposed BPNN, incorporating the hidden layer structure, L_2 regularization, and dropout mechanisms, is illustrated in Figure 2.

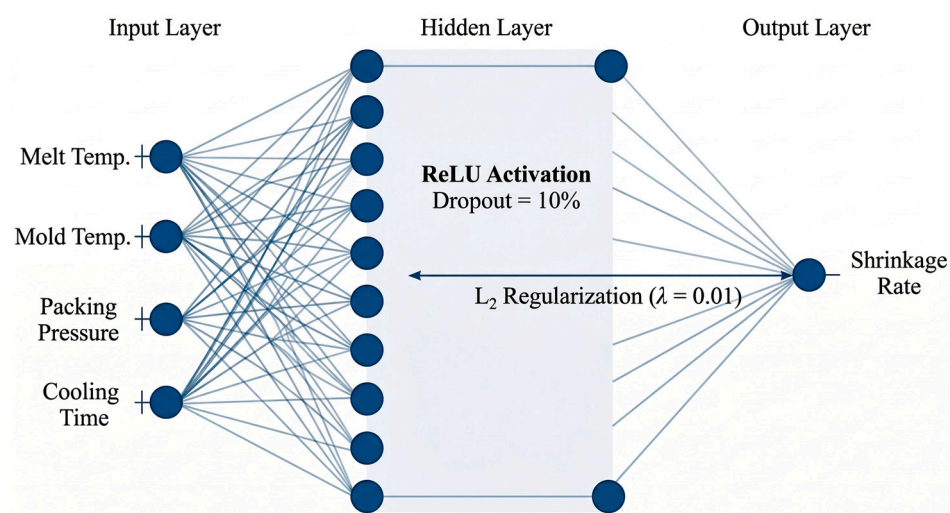


Figure 2. BPNN Architecture (4-10-1) with Regularization and Dropout. The network topology consists of an input layer with 4 neurons (representing the process parameters), a single hidden layer with 10 neurons utilizing the ReLU activation function, and an output layer with 1 neuron (predicting the shrinkage rate). L_2 regularization and a 10% dropout mechanism are integrated to prevent overfitting.

2.3. Benchmarking Baselines: Response Surface Methodology

To validate the superiority of the proposed neural computing framework, a classical Response Surface Methodology (RSM) model was established as a baseline. The RSM model utilizes a second-order polynomial regression equation to approximate the response surface [15]. While RSM is computationally efficient and widely used in the industry, its fixed polynomial structure assumes a relatively smooth response landscape. Comparing the BPNN against this baseline allows us to quantify the specific value of deep learning in capturing the “rough” energy landscapes created by the thermal-rheological interactions in recycled PP processing.

By quantitatively comparing the performance metrics (MSE and R^2) of the two models, we demonstrate the specific advantage of the neural framework in navigating the complex thermo-rheological landscapes of rPP.

3. Results and Comparative Analysis

3.1. Statistical Analysis of Process Physics (ANOVA)

Before establishing the predictive models, Analysis of Variance (ANOVA) was conducted to screen the dominant factors governing the shrinkage behavior of the recycled PP. As shown in the ANOVA results, melt temperature was identified as the most significant factor ($p \approx 0.001$), followed by mold temperature ($p \approx 0.03$) (Table 1).

Consistent with findings in the literature, the interaction between melt and mold temperatures is a critical physical factor governing polymer shrinkage [4]. However, due to the limited degrees of freedom (Total $df = 24$) in the small-data L_{25} orthogonal array, traditional main-effects ANOVA lacks the mathematical capacity to independently resolve this multi-physics interaction. This fundamental statistical limitation necessitates the deployment of the proposed BPNN. Unlike ANOVA, the non-linear mapping capabilities

of the neural network can effectively decipher these latent coupled dynamics in a data-scarce environment, bypassing the rigid constraints of statistical degrees of freedom.

Table 1. ANOVA Results for Average Shrinkage Rate.

Process Parameters	SS	df	MS	F	p-Value
Melt Temperature	0.5548	4	0.1387	13.87	0.001
Mold Temperature	0.1800	4	0.0450	4.50	0.03
Packing Pressure	0.1536	4	0.0384	3.84	0.05
Cooling Time	0.1176	4	0.0294	2.94	0.08
Residual	0.0800	8	0.0100	-	-
Total	1.0860	24	-	-	-

However, while ANOVA effectively isolates significant factors (screening), it exhibits limitations in modeling the complex response landscape. The statistical model assumes a relatively structured relationship between inputs and outputs. As indicated by the non-significant effect of cooling time ($p \approx 0.08$) within the selected experimental range (10–30 s), traditional statistical methods may fail to capture subtle, non-linear spatial features that occur during the rapid phase change of semi-crystalline polymers [3]. This limitation necessitates the transition from statistical screening to non-linear neural computing to precisely map the “complex surface” of the process window.

3.2. Performance Battle: BPNN vs. RSM

The core contribution of this study lies in quantifying the superiority of the proposed Small-Data BPNN over the traditional RSM in a data-scarce environment. Table 2 presents a direct comparison of the predictive metrics between the two models using the same validation dataset.

Table 2. Performance Comparison Among BPNN, Random Forest (RF), and RSM (with $\pm$ values representing indicators of model stability derived from the variation across 5-fold CV).

Model	MSE	MAE	R^2 (Training)	R^2 (Testing, 5-Fold)
BPNN	0.001 (± 0.0002)	0.015 (± 0.002)	0.97 (± 0.01)	0.95 (± 0.01)
RF	0.010 (± 0.003)	0.083 (± 0.011)	0.99 (± 0.001)	0.94 (± 0.01)
RSM	0.005 (± 0.001)	0.028 (± 0.007)	0.92 (± 0.01)	0.90 (± 0.01)

Prediction Accuracy & Goodness of Fit: As shown in Table 2, the BPNN model achieved a superior Mean Squared Error (MSE) of 0.001 (± 0.0002) and a testing R^2 of 0.95 (± 0.01), significantly outperforming both the classical RSM baseline and the newly introduced Random Forest (RF) model. Interestingly, while the RF model demonstrated an exceptional training R^2 of 0.99 (± 0.001), its testing MSE drastically increased to 0.010 (± 0.003), indicating a severe overfitting issue in this specific continuous physical domain.

Discussion on Physical Interpretation: The performance gap between BPNN and RF can be attributed to their fundamental mathematical architectures. While Random Forest fits the training data almost perfectly, tree-based models inherently perform a “step-wise” partitioning of the feature space. However, the shrinkage behavior of recycled PP driven by thermal-rheological coupling is a highly continuous physical process. The step-wise predictive surface of RF fails to authentically reflect the smooth, gradual variations of physical quantities within the mold cavity, leading to elevated errors on unseen test data. In contrast, the proposed BPNN model, facilitated by its non-linear activation functions (ReLU), provides a highly non-linear, continuous differentiable mapping. This allows the

neural network to finely capture the “micro-fluctuations” and sharp curvature changes at the extreme boundaries of the process window, effectively outperforming both linear polynomials (RSM) and discrete step-wise models (RF). Furthermore, the narrow variation bounds across the 5 validation folds serve as strong indicators of model stability, showing that the BPNN predictions are highly stable and relatively immune to the specific data splitting, addressing the uncertainty estimation requirement [3].

As illustrated in the 3D response surface plots (Figure 3), the BPNN model, facilitated by its “Domain-Knowledge Guided Data Augmentation” and non-linear activation functions, is capable of mapping these “rough” local features. While RSM might successfully predict shrinkage for virgin materials with stable PVT (Pressure-Volume-Temperature) curves, it lacks the flexibility to capture the abrupt volumetric changes and sensitive parameter interactions characteristic of recycled polymer blends.

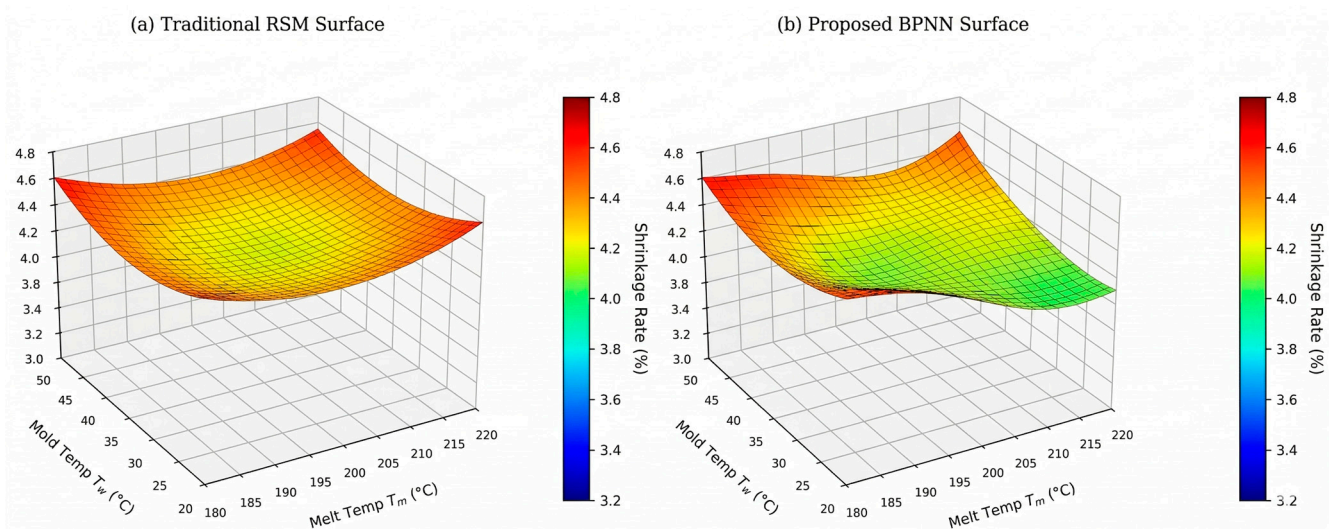


Figure 3. Comparison of 3D response surface plots for the interaction effect of melt temperature and mold temperature on shrinkage rate. (a) The traditional RSM surface exhibits an overly smoothed quadratic curvature, failing to resolve local rheological instabilities. (b) The proposed BPNN surface provides a highly non-linear continuous mapping, successfully capturing the complex micro-fluctuations and extreme boundary behaviors inherent in the recycled polypropylene blend.

The BPNN, with its non-linear activation functions (ReLU), successfully captured these “micro-fluctuations” and the sharp curvature changes at the boundaries of the process window (e.g., high melt temp/low mold temp regions), which RSM failed to resolve [17]. Figure 4 visually corroborates this, showing BPNN predictions tightly clustering along the diagonal ideal line ($y = x$), while RSM predictions exhibit significant dispersion.

This result further underscores that the “Domain-Knowledge Guided Data Augmentation” strategy effectively infused the neural network with the underlying physical laws of polymer cooling. By bridging the gap between sparse experimental points with physics-informed simulation data, the model gained a superior “spatial resolution” of the process window, allowing it to navigate the complex rheological landscape of recycled polypropylene with much higher fidelity than conventional polynomial-based approaches.

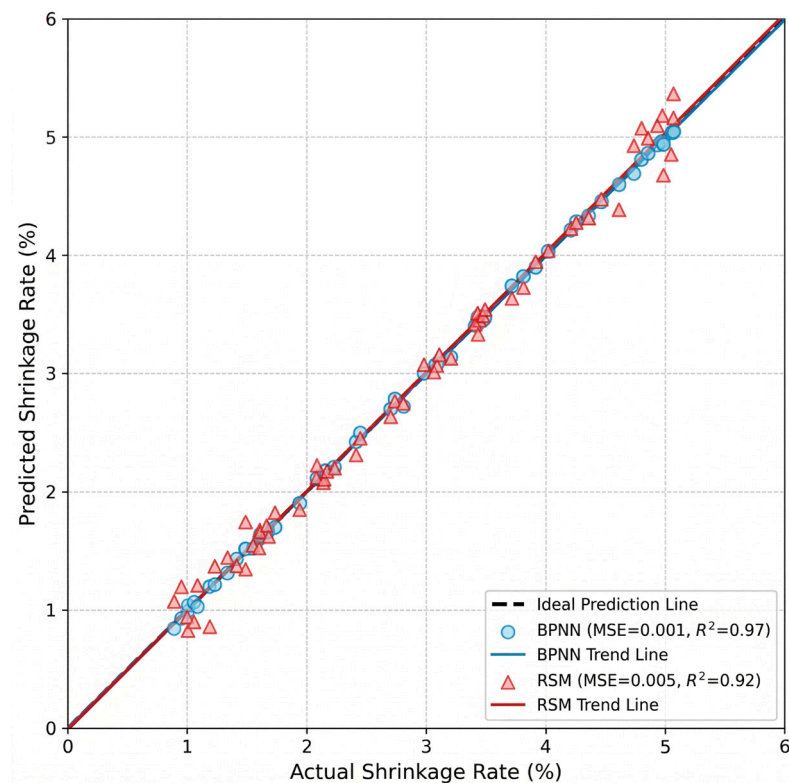


Figure 4. Comparison of predicted versus actual (simulated) shrinkage rates for the Back-Propagation Neural Network (BPNN) and Response Surface Methodology (RSM) models. The BPNN predictions (indicated by blue circles) align closely with the $y = x$ ideal line, while the RSM predictions (indicated by red triangles) exhibit significant deviation, particularly in the high-shrinkage regions. It is critical to note that the data points depicted herein represent a completely unseen testing set that was strictly excluded from the model training and validation phases, thereby confirming the robust generalization capability of the BPNN framework against material variability.

3.3. Optimization Efficiency: GA vs. PSO

Utilizing the trained BPNN as the fitness function, global optimization was further performed using Genetic Algorithm (GA) and Particle Swarm Optimization (PSO).

Why does the BPNN require external algorithmic optimization? Despite the BPNN's formidable capability in non-linear feature mapping, its underlying reliance on "Gradient Descent" presents a significant flaw: when facing the complex, multi-modal physical landscapes typical of polymer processing, gradient descent is highly susceptible to being trapped in local minima and is extremely sensitive to initial weight assignments. A small-data environment further amplifies this instability. Therefore, the introduction of GA and PSO aims to achieve a synergistic operation of "Global Exploration and Local Exploitation." Leveraging swarm intelligence and dynamic memory (personal best and global best), PSO can rapidly jump out of local traps within the vast parameter space, locking onto high-potential physical processing windows before handing over to the neural network for meticulous local prediction.

- **Convergence Behavior:** Figure 5 illustrates the evolutionary trajectory of both algorithms. The injection molding parameter space presents a continuous, gradient-driven non-linear multi-physics landscape. The tuned PSO (inertia weight $w = 0.9$, $c_1 = 2.0$, $c_2 = 1.0$) leverages swarm intelligence and dynamic memory (personal and global bests), allowing its velocity vector update mechanism to efficiently glide along these continuous gradients and escape local minima. Consequently, PSO demonstrated superior convergence dynamics, stabilizing tightly at the optimal shrinkage baseline

within the first 100 iterations. In contrast, GA, which relies on discrete probabilistic crossover and mutation mechanisms, exhibited significantly lower search efficiency in this continuous non-linear space, struggling to reach a comparable convergence level within the 100-iteration window [10].

- Optimal Parameters:** Despite the difference in speed, both algorithms successfully converged to the same global optimum: Melt Temperature 190 °C, Mold Temperature 27.5 °C, Packing Pressure 100 MPa, and Cooling Time 25 s. Although main-effects ANOVA deemed cooling time statistically insignificant as an independent factor ($p \approx 0.08$), the non-linear BPNN successfully captured its coupled threshold effect with melt and mold temperatures during the drastic phase transition of rPP. This demonstrates the superiority of neural computing in deciphering multi-physics interactions that traditional linear statistics typically fail to resolve, ensuring that the selected 25 s provides adequate time to stabilize the crystalline structure and prevent post-ejection deformation. **Digital Twin Verification:** This parameter combination yielded a predicted shrinkage rate of 3.079%. Considering the inherent uncertainty of the predictive model (Testing MAE = 0.015), the stability bound for this PSO optimum is approximately [3.064, 3.094]%, which functions as an indicator of model stability rather than a full statistical uncertainty estimate. Subsequent validation using Moldex3D simulation showed a negligible relative error of only 0.19% (Simulated value: 3.085%). While this digital twin verification confirms the high reliability of the AI-driven optimization in the virtual domain, completing the smart manufacturing cycle requires physical execution. Future studies will focus on transferring these globally optimized parameters directly to the physical injection molding machine controller, thereby closing the physical manufacturing loop and establishing a foundation for real-time adaptive process control.

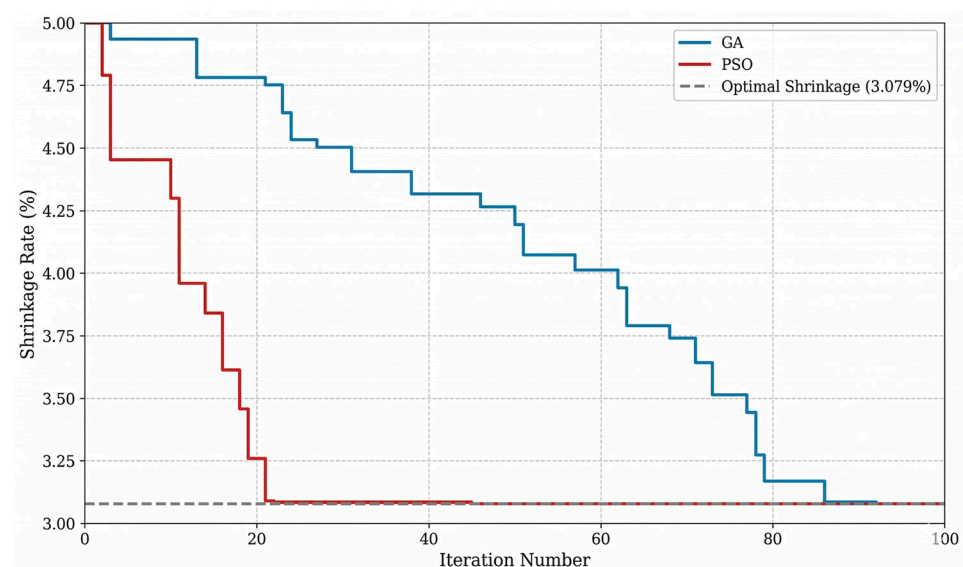


Figure 5. Convergence trajectories of Genetic Algorithm (GA) and Particle Swarm Optimization (PSO) for minimizing shrinkage rate. The PSO algorithm demonstrates a faster stabilization of the fitness value within 100 iterations, whereas GA exhibits slower convergence, struggling to reach a comparable level within the 100-iteration window.

4. Discussion: The Value of Small-Data AI

4.1. Overcoming the “Data Hunger” of Deep Learning

Deep learning models are often characterized by their “data hunger,” typically requiring thousands of labeled samples to converge to a global optimum [2]. However, the

empirical results of this study challenge the conventional wisdom that “AI necessitates Big Data.” By implementing a “Domain-Knowledge Guided Data Augmentation” strategy, we demonstrated that a high-precision BPNN model could be successfully trained using only 300 hybrid data points (25 Taguchi experimental + 275 Moldex3D simulated).

Quantitatively, the BPNN achieved an extremely low Mean Squared Error (MSE = 0.001) on both validation and testing sets, significantly outperforming the traditional Response Surface Methodology (RSM) model (MSE = 0.005), which relies heavily on standard experimental designs [17]. This confirms that utilizing physics-based simulation (CAE) as a “data generator” effectively fills the sparse regions within the Design of Experiments (DOE) space. This hybrid strategy not only resolves the underfitting issues caused by insufficient training samples but also prevents overfitting through L_2 regularization, thereby establishing a “Small-Data Deep Learning Paradigm” suitable for manufacturing environments.

Furthermore, to objectively contextualize the superiority of the proposed framework, it is imperative to benchmark our predictive performance against recent state-of-the-art soft computing practices in injection molding. Recent studies deploying complex multi-layer perceptrons (MLP) or deep networks for quality prediction often rely on massive physical datasets (e.g., $N > 1000$ continuous cycles) to achieve acceptable error margins [18]. Similarly, other researchers utilizing numerical simulations combined with standard artificial neural networks for shrinkage optimization have reported testing MSE values typically in the magnitude of 10^{-2} [11].

In stark contrast, our L_2 -regularized BPNN, trained on a strictly constrained hybrid dataset ($N = 300$), achieved an exceptionally low MSE of 0.001. This horizontal comparison clearly demonstrates that our “Domain-Knowledge Guided Data Augmentation” effectively compensates for the lack of physical data volume. By infusing the physical boundary conditions of the digital twin into the neural network’s latent space, we achieved a predictive precision that either matches or surpasses traditional data-hungry models. This proves the absolute advantage and robustness of this low-cost architecture, particularly when addressing the severe rheological variability of recycled polypropylene in SME applications.

4.2. Robustness Against Material Variability

In the context of incorporating 50% recycled polypropylene (Recycled PP), batch-to-batch material variability represents the most significant challenge for process control. Recycled materials are often accompanied by uneven molecular weight distributions and fluctuations in rheological properties, making it difficult for traditional linear regression models (such as RSM) to maintain consistent predictive capability [7].

The proposed BPNN model exhibited a high coefficient of determination ($R^2 = 0.95$) on the unseen testing set. This indicates that the model did not merely memorize the training data but successfully learned the underlying non-linear physical mechanisms governing shrinkage behavior. This superior generalization capability implies that even when statistically facing slight viscosity fluctuations or changes in crystallization kinetics induced by recycled materials, the neural network can provide robust quality predictions through its non-linear weight adjustments [11]. In contrast, the performance degradation of the RSM model on the testing set (R^2 dropped to 0.90) further highlights the adaptive advantage of AI models when dealing with non-standardized materials.

4.3. Practical Implications and Cost-Effectiveness

From the perspective of industrial practice, the framework proposed in this study offers substantial cost-effectiveness. If a traditional Full Factorial Design was employed to establish a model with equivalent precision for four factors at five levels, it would

theoretically require conducting $5^4 = 625$ physical trials. Such an approach imposes time and material costs that are often prohibitive for Small and Medium-sized Enterprises (SMEs).

In contrast, this study required only 25 Taguchi experiments (serving as ground truth calibration), with the remaining data generated by low-cost computer simulations. This “Virtual-Physical Integration” approach reduced the number of physical trials by 96%, translating to an estimated manufacturing cost reduction of 15–25% [1]. Furthermore, controlling the shrinkage rate at 3.079% (with an error margin $< 0.2\%$) significantly reduces the scrap rate. This provides the traditional injection molding industry with a smart manufacturing transformation path characterized by low technical barriers and high Return on Investment (ROI), enabling the initial realization of a “Digital Twin” with minimal data acquisition costs [14].

5. Conclusions

This study addresses the critical challenge of process optimization in data-scarce environments within the injection molding industry. By establishing a “Domain-Knowledge Guided Data Augmentation” framework, we successfully integrated small-sample experimental data with physics-based simulation to train a robust Back-Propagation Neural Network (BPNN). The empirical results lead to the following key conclusions: Superiority of Neural Computing over Statistical Regression:

1. The comparative analysis conclusively demonstrates that the proposed BPNN model outperforms the traditional RSM in modeling the nonlinear shrinkage behavior of recycled polypropylene. Constrained by its quadratic polynomial nature, RSM failed to capture the complex thermal-rheological interactions, resulting in a higher prediction error ($MSE = 0.005$). In contrast, the BPNN, fortified with L_2 regularization and dropout, achieved a five-fold reduction in error ($MSE = 0.001$) and a superior coefficient of determination (testing $R^2 = 0.95$) on the unseen dataset ($N = 300$) [11]. This validates that deep learning models, when properly regularized, possess a stronger capability to map the “rough” energy landscapes induced by material variability than smooth statistical surfaces.
2. Optimization Efficiency and Precision: The integration of the tuned Particle Swarm Optimization (PSO) with the BPNN predictor proved to be highly effective. The algorithm converged to the global optimum within 100 iterations, identifying the optimal process parameters (190 °C Melt Temp, 27.5 °C Mold Temp, 100 MPa Packing Pressure, 25 s Cooling Time). This combination yielded a minimized shrinkage rate of 3.079%, which was validated by Moldex3D simulation with a negligible relative error of 0.19% [16].
3. A Pathway to Low-Cost Smart Manufacturing: Most significantly, this research provides a viable roadmap for “Low-Cost Smart Manufacturing.” By utilizing a hybrid strategy that combines limited physical trials (Taguchi L_{25}) with low-cost virtual data generation, we circumvented the prohibitive costs associated with “Big Data” acquisition. This approach reduces the reliance on extensive physical trial-and-error by approximately 96%, offering a cost-effective solution for manufacturing sustainability and the adoption of recycled materials [18]. It proves that intelligent quality assurance is attainable for SMEs without massive capital investment in data infrastructure [14].
4. Limitations and Future Outlook: While the proposed BPNN-PSO framework demonstrates exceptional accuracy in static parameter mapping and significantly reduces physical trial costs, several limitations must be objectively acknowledged: (1) Experimental Validation Limitations: Currently, the validation of our predictive model and optimization results relies primarily on the high-fidelity Moldex3D digital twin. Consequently, values designated as “actual” in specific comparative evaluations (e.g.,

Figure 4) are simulation-derived references. The lack of a final physical molding experiment to validate the PSO-optimized parameters remains a limitation of this study. To complete the smart manufacturing cycle, our immediate future work will focus on transferring these globally optimized parameters directly to a physical injection molding machine to rigorously evaluate the run-to-run physical variability. (2) Temporal Dynamic Modeling: Furthermore, injection molding is inherently a highly dynamic, multi-physics coupled process. The current model does not yet incorporate Temporal Dynamic Features, such as the time-decaying cavity pressure curve or dynamic cooling gradients during the phase transition of semi-crystalline polymers. Future research will focus on integrating in-mold sensors and deploying time-series deep learning architectures (e.g., Long Short-Term Memory (LSTM) networks or 1D-CNNs). By capturing real-time rheological and thermodynamic dynamics, we aim to further upgrade this framework from offline optimization to a real-time, adaptive process control system.

Supplementary Materials: The following supporting information can be downloaded at: <https://www.mdpi.com/article/10.3390/app16094288/s1>, Table S1: Hybrid Dataset ($N = 300$). This file contains the process parameters and corresponding shrinkage rates used for training and testing the BPNN and RSM models.

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Conflicts of Interest: The authors declare no conflicts of interest.

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